Project Proposal

Friday, September 11, 2026 9:31 AM

Effect of ASR Errors on Spoken Dialog System Performance for Speakers with Dysarthria

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September 2026

1. Summary

We will build two cascaded spoken dialog systems (streaming ASR → LLM → TTS) that are identical in every respect except the ASR checkpoint — one using Voxtral Realtime as released, the other using Voxtral Realtime fine-tuned on SAP data — and measure how the difference in ASR affects *dialog task outcomes* for speakers with dysarthria. The systems will be tested first by offline replay of SAP recordings, then in an internal pilot, then by UIUC students or community members with dysarthria. The project answers a question the SAP partner companies have asked directly: what does an SAP-fine-tuned ASR buy at the level of a working assistant, not just at the level of WER?

2. Background and Motivation

Fine-tuning ASR on SAP data lowers WER on dysarthric speech; this is established. What is not established is how much of that improvement survives the rest of a dialog system. ASR errors propagate into the LLM, which may compensate (ask a clarifying question, infer intent from context) or fail (execute the wrong action, hallucinate a request). Our earlier work on the SAP-Hypo5 benchmark showed that WER is a poor proxy for downstream usefulness on dysarthric speech and that LLM-based processing can recover meaning from noisy hypotheses. The natural next step is to close the loop: put a real streaming ASR in front of a real LLM and TTS, and measure task success, efficiency, and user experience with the intended population.

A second issue is specific to this population and largely absent from mainstream voice-assistant engineering. Speakers with dysarthria have slower speech rate, longer within-utterance pauses, and slower speech onset. Silence-based end-of-turn detection, the default in production systems, cuts them off before ASR has a chance to succeed or fail. Turn-taking is therefore both a confound to control in the ASR study and a research direction in its own right.

3. Research Questions and Hypotheses

RQ1 (primary). Does SAP fine-tuning of a streaming ASR improve dialog task success, efficiency (turns and time to completion), and user-rated experience for speakers with dysarthria? *H1:* Condition B (fine-tuned) achieves higher task success and fewer clarification turns than Condition A (base), and the improvement is larger than the WER improvement alone would predict for some tasks and smaller for others, depending on which words are misrecognized.

RQ2. Which ASR error types cause dialog failure, and to what extent does the LLM compensate? *H2:* Content-word substitutions in slot positions (names, times, quantities) drive most failures; dropped function words and disfluency insertions are largely absorbed by the LLM.

RQ3 (stretch). How does model-based turn detection behave on dysarthric speech, and can it be adapted with SAP data? *H3:* Off-the-shelf semantic end-of-turn models degrade on dysarthric speech but degrade less than silence-threshold VAD, and fine-tuning on SAP spontaneous speech recovers most of the gap.

4. Approach

System. A cascaded, pipelined spoken dialog system: microphone → push-to-talk turn control → Voxtral Realtime (streaming) → LLM with

a fixed task-assistant prompt → sentence-level chunking → TTS → speaker. Every stage runs concurrently with the next, so response latency is on the order of one second rather than several. Each stage sits behind a fixed streaming interface, so components can be swapped by configuration.

Two conditions, one variable.

- Condition A: Voxtral Realtime, base checkpoint.
- Condition B: Voxtral Realtime, fine-tuned on SAP training data (speaker-disjoint from all evaluation data). LLM, prompt, TTS voice, turn control, latency pipeline, UI, and logging are identical across conditions.

Why a cascade rather than end-to-end speech-to-speech. The question is about ASR errors, which are only observable in a cascade. A cascade also lets the partner companies swap in their own ASR, LLM, or TTS against the same harness.

Why push-to-talk. It removes turn-detection failures from the A/B comparison so that any difference is attributable to ASR. It is implemented as tap-to-start / tap-to-stop rather than press-and-hold, since dysarthria frequently co-occurs with motor impairment. Model-based turn detection is studied separately and, if resources allow, becomes a second experimental axis (RQ3).

Component studies. Two undergraduate-led studies inform the fixed choices and give the partners reusable results: (1) LLM robustness versus latency on the actual transcripts produced by the base and the SAP-fine-tuned Voxtral Realtime on SAP test audio — i.e., on the two error distributions the final system will see, which also shows how much of the ASR improvement each LLM can already compensate for; (2) TTS latency and intelligibility under streaming sentence-level input, plus a comparison of silence-threshold VAD, neural VAD, and the X2-Turn semantic turn-state model on dysarthric SAP audio.

5. Evaluation

Offline replay. SAP test recordings are played through both pipelines. Reported: WER, intent accuracy, slot accuracy, clarification rate, LLM-judged response quality. This produces a result independent of participant recruitment.

Internal pilot. Lab members, including deliberately slow speech, to debug the protocol and confirm session length.

Participant pilot. Two to three participants with dysarthria. Within-subject design, condition order counterbalanced, six scripted task scenarios with explicit success criteria (set an alarm, check weather, order a meal with modifications, reschedule an appointment, add a calendar event, ask for directions), short questionnaire after each condition.

Metrics. ASR: live WER per task. Dialog: task success, turns to completion, clarification requests, time to completion. System: end-to-end and per-stage latency as P50/P95/P99. User: perceived effort, satisfaction, preference. All per-turn audio, transcripts, LLM I/O, and timestamps are logged.

6. Deliverables to Partners

1. Two runnable systems (A and B) with frozen configurations and checkpoint identifiers, and a harness that accepts alternative ASR/LLM/TTS components.
2. WER results on SAP test data for base vs. fine-tuned Voxtral Realtime, with confidence intervals.
3. Offline replay results and participant pilot results, with an error analysis linking ASR error types to dialog failures.
4. Component reports: LLM robustness-vs-latency, TTS latency/intelligibility, turn detection on dysarthric speech.
5. A technical report with recommendations.

7. Timeline

Phased, roughly eight weeks to a participant-tested system.

Phase	Approx. duration	Content
0	2–3 days	Verify streaming path accepts custom checkpoints; IRB inquiry; skeleton choice
1	~3 weeks	Core framework, fine-tuning, task design; undergrads read and prepare study designs
2	~3 weeks	Component studies, offline replay, internal pilot
3	~2 weeks	System freeze, participant pilot, report
4	open	Turn-taking as a second experimental axis; SAP-adapted turn detector

8. Ethics, IRB, and Privacy

Participant testing requires either coverage under the existing SAP protocol or an amendment; the inquiry is sent on the first day of the project. If an API-hosted LLM or TTS is used, participant transcripts leave the lab and the consent form will say so; a locally hosted LLM avoids this. Audio is captured raw (no noise suppression or gain control) and stored with anonymized participant identifiers.

9. Stretch Direction

Turn detection for dysarthric speech has not been studied. X2-Turn (a frame-synchronous ASR + turn-state model built on the same Voxtral Realtime architecture we fine-tune) makes it feasible to (a) compare push-to-talk against model-based turn-taking in a 2 × 2 design with the ASR axis, and (b) fine-tune the turn-state head on SAP spontaneous speech.